EU Declaration of Conformity

EU Declaration of Conformity

According to

EMC Directive 2004/108/EC

For the following

Product	:	**Analog IO Board Set**
Model Name	:	**SIO-AI8F V1.0, SIO-AO4F V1.0, SIO-AI8AO4F V1.0**
Manufactured at	:	**AJINEXTEK CO.,LTD.**
Address	:	**9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea**

We hereby declare, Electromagnetic Compatibility Directives (2004/108/EC) are fulfilled, as laid out in the guideline set down by the member states of the EEC Commission. This declaration is valid for all samples that are part of this declaration, which are manufactured according to the production charts appendix.

The standards relevant for the evaluation of EMC requirements are as follows:

Test Standards	:	**EN 55022:2010, Class A**
		EN 55024: 2010
		EN 61000-3-2:2006+A2:2009
		EN 61000-3-3:2008

Date of issue: September 05, 2013

AJINEXTEK CO.,LTD.

9-3 Horim-dong, Dalseo-gu,
Daegu 704-240, Korea

(Name and signature of authorized person)

EMC TEST REPORT

Test report No	:	**EMC-CE-4270**
Type of Equipment	:	**Analog IO Board Set**
Model Name	:	**SIO-AI8F V1.0, SIO-AO4F V1.0,**
		SIO-AI8AO4F V1.0
Applicant	:	**AJINEXTEK CO.,LTD.**
		9-3 Horim-dong, Dalseo-gu,
		Daegu 704-240, Korea
Manufacturer	:	**AJINEXTEK CO.,LTD.**
		9-3 Horim-dong, Dalseo-gu,
		Daegu 704-240, Korea
Test standards	:	**EN 55022:2010, Class A**
		EN 55024:2010
		EN 61000-3-2:2006+A2 :2009
		EN 61000-3-3:2008
Testing Laboratory	:	**EMC Compliance Ltd.**
Test result	:	**Complied**

This product complies with the requirements of the EMC Directive 2004/108/ EC.
The results in this report apply only to the sample tested.
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Date of receipt: 2013. 08. 22

Date of testing: 2013. 08. 30 ~ 09. 02

Issued date: 2013. 09. 05

Tested by: LEE, MIN-CHANG

Approved by: YEOM, HAN-SEOK

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1. Applicant information

Applicant:	AJINEXTEK CO.,LTD.
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Telephone :	+82-53-593-3700
Fax:	+82-53-593-3703
E-mail:	queive@ajinextek.com
Contact name:	**Jeon Jong Hoon**

Manufacturer:	AJINEXTEK CO.,LTD.
Address:	9-3 Horim-dong, Dalseo-gu, Daegu 704-240, Korea
Telephone :	+82-53-593-3700
Fax:	+82-53-593-3703
E-mail:	queive@ajinextek.com
Contact name:	**Jeon Jong Hoon**

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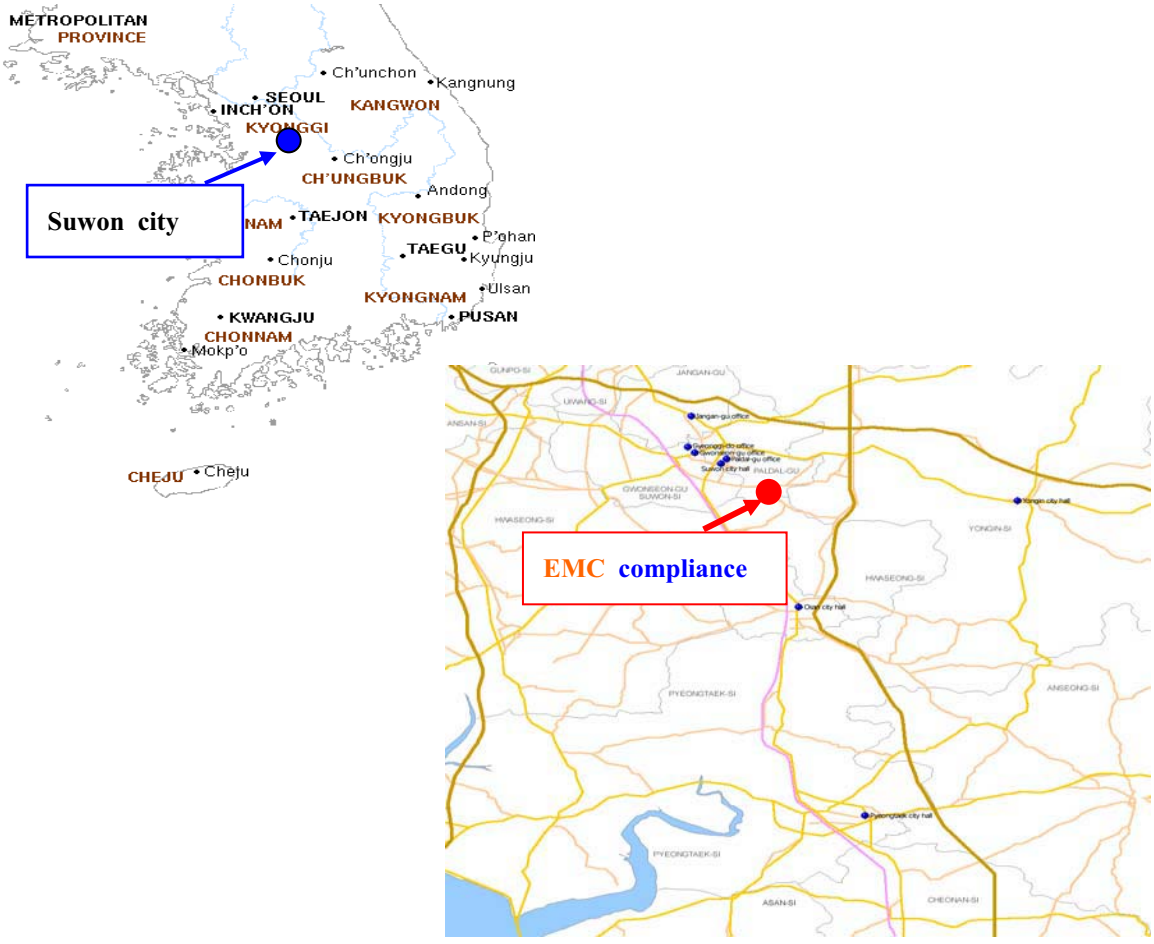
2. Laboratory information

Address

EMC compliance Ltd.
480-5 Sin-dong, Yeongtong-gu, Suwon-city, Gyeonggi-do, 443-390, Korea
Telephone Number: 82 31 336 9919
Facsimile Number: 82 505 299 8311

FCC CAB.: KR0040
VCCI Registration No. : R-3327, G-198, C-3706, T-1849
Industry Canada Registration No. : 8035A
KOLAS NO.: 231

SITE MAP



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3. Test system configuration

3.1 Operation environment

		Temperature	Humidity	Pressure
Chamber(10 m)	:	25 °C	50 % R.H.	-
Shielded room(CE)	:	25 °C	42 % R.H.	-
Shielded room(ESD)	:	25 °C	45 % R.H.	99.7 kPa

Test site

These testing items were performed following locations;

Test item	Test site
Conducted Emission	Shielded Room
Radiated Emission	10 m Chamber
Harmonics current	Immunity area
Voltage fluctuations and flickers	Immunity area
Electrostatic discharge	Shielded Room
Radiated RF immunity	Fully anechoic chamber (3 m)
Electric Fast Transient/BURST	Shielded Room
Surge	Shielded Room
Conducted RF immunity	Shielded Room
Magnetic field immunity	Shielded Room
Voltage dip/interruption	Shielded Room

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3.2 Measurement Uncertainty

All measurements involve certain levels of uncertainties, especially in field of EMC.
The factors contributing to uncertainties are test receiver, cable loss, antenna factor calibration, Antenna directivity, antenna factor variation with height, antenna phase center variation, antenna frequency interpolation, measurement distance variation, site imperfection, mismatch, and system repeatability. Based on CISPR 16-4-2, the measurement uncertainty level with a 95 % confidence level was applied.

Conducted emission measurement (C.L: Approx 95 %, k = 2)		
Shielded Room (CE#1)	9 ㎑ ~ 150 ㎑: ± 3.82 dB 150 ㎑ ~ 30 ㎒: ± 3.43 dB	
Shielded Room (CE#2)	9 ㎑ ~ 150 ㎑: ± 3.82 dB 150 ㎑ ~ 30 ㎒: ± 3.43 dB	
Shielded Room (CE#3)	9 ㎑ ~ 150 ㎑: ± 4.00 dB 150 ㎑ ~ 30 ㎒: ± 3.63 dB	
Radiated Emission measurement (C.L: Approx 95 %, k = 2)		
10 m Chamber (#F4)	30 ㎒ ~ 300 ㎒	3 m: + 4.56 dB, - 4.58 dB 10 m: + 4.56 dB, - 4.56 dB
	300 ㎒ ~ 1 000 ㎒	3 m: + 4.84 dB, - 4.85 dB 10 m: + 4.71 dB, - 4.72 dB
	1 ㎓ ~ 6 ㎓	3 m: + 6.19 dB, - 6.20 dB
10 m Chamber (#F2)	30 ㎒ ~ 300 ㎒	3 m: + 4.86 dB, - 4.88 dB 10 m: + 4.86 dB, - 4.86 dB
	300 ㎒ ~ 1 000 ㎒	3 m: + 4.98 dB, - 4.99 dB 10 m: + 4.85 dB, - 4.87 dB
	1 ㎓ ~ 6 ㎓	3 m: + 6.19 dB, - 6.20 dB
Radio Frequency Electromagnetic Fields (C.L: Approx 95 %, k = 2)		
± 1.82 dB		
Disturbance power Electromagnetic Fields (C.L: Approx 95 %, k = 2)		
± 3.73 dB		

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4. Description of E.U.T.

4.1 General information

* SIO-AI8F V1.0

Items	Specifications
Number of channels	8 Channel
Resolution	16Bit
Voltage range	±10V, ±5V, 4~20mA (Current Mode)
Mode	Single-ended Mode
Sampling rate	Use Input 1 channel up to 40us
Precision	Full Scale @ ± 0.02% or less
Trigger Mode	Software
Internal current consumption	Max.210mA(5V), Max.35mA(3.3V)
Isolation	Input / output terminals and power isolation between
Operating temperature / humidity	Temperature 0 ~ 60 ° C / Humidity 5-90% (non-condensing)
Storage Temperature	-20 ~ 70°C
Weight	30g
Board Dimensions	120 (W) mm x 45 (L) mm

* SIO-AO4F V1.0

Items	Specifications
Number of channels	4 Channel
Resolution	16Bit
Voltage range	±10V, ±5V
Mode	Voltage Mode
Data set-up time	Output when using 1channel up 4us (Settling Time : Max 12us)
Precision	Full Scale @ ±0.04%이하
Internal current consumption	Max.280mA(5V), Max.35mA(3.3V)
Isolation	Isolation between the output terminal and the power
Operating temperature / humidity	Temperature 0 ~ 60 ° C / Humidity 5-90% (non-condensing)
Storage Temperature	-20 ~ 70°C
Weight	30g
Board Dimensions	120 (W) mm X 45 (L) mm

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* SIO-AI8AO4F V1.0

Items	Specifications
Number of channels	Input 8 Channel, Output 4 Channel
Resolution	Input / Output 16Bit
Voltage range	Input ±10V, ±5V, 4~20mA / Output ±10V, ±5V
Mode	Single-ended Mode
Sampling rate	Use Input 1 channel up to 40us
Data set-up time	Output when using 1channel up 4us (Settling Time : Max 12us)
Precision	Less than ± 0.02% Full Scale @ input / output less than ± 0.04%
Trigger Mode	Software (Analog input only)
Internal current consumption	Max.350mA(5V), Max.40mA(3.3V)
Isolation	Input / output terminals and power isolation between
Operating temperature / humidity	Temperature 0 ~ 60 ° C / Humidity 5-90% (non-condensing)
Storage Temperature	-20 ~ 70°C
Weight	30g
Board Dimensions	120 (W) mm x 45 (L) mm

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4.2 Product description

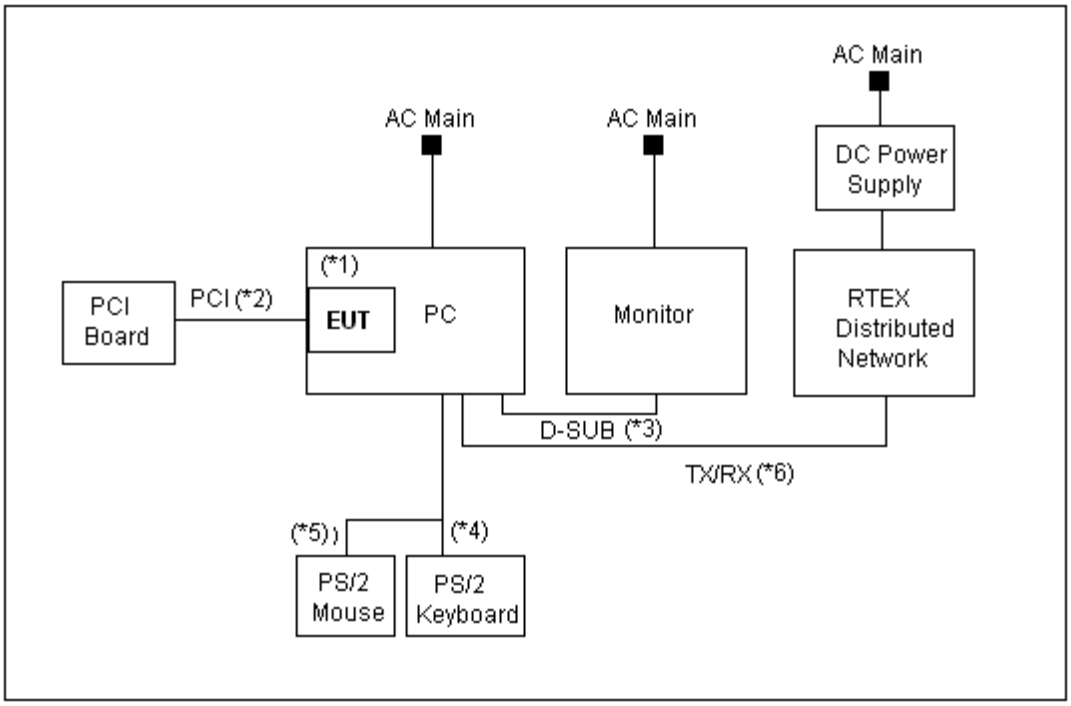
Type of product	Analog IO Board Set
Model name (Basic)	SIO-AI8F V1.0, SIO-AO4F V1.0, SIO-AI8AO4F V1.0
Model name (Variant)	-
Difference	-
Trade name	-
Serial no	Engineering Sample
Testing voltage	230 Ⅴ, 50 ㎐
Product rating	230 Ⅴ, 50 ㎐
Internal clock frequency	50 ㎒
Note	-

4.3 Auxiliary equipments

Type	Model / Part #	Serial number	Manufacturer
RTEX Distributed Network	RTEX-AI16F	-	AJINEXTEK CO.,LTD.
PC	-	-	-
Monitor	PM24KO	ZQ9XH1HBB00255L	SAMSUNG
PS/2 Mouse	-	-	SAMSUNG
PS/2 Keyboard	SEM-DT35	44000737	SAMSUNG
PCI Board	-	-	-
DC Power Supply	-	-	-

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4.4 Test configuration



Note	Start		End		Cable	
*	Name	I/O port	Name	I/O port	Length (m)	Spec.
1	**EUT** (Analog IO Board Set)	-	PC	-	Direct	-
2		PCI	PCI Board	PCI	3.0	Shield
3	PC	D-SUB	Monitor	D-SUB	1.8	Shield
4		PS/2	PS/2 Keyboard	PS/2	2.0	Non-Shield
5			PS/2 Mouse	PS/2	2.0	Non-Shield
6		TX/RX	RTEX Distributed Network	TX/RX	3.0	Non-Shield

4.5 Operating conditions

The EUT was configured as normal intended use.

Test mode	Normal operating
1	Analog IO Board Set operating test. (Program: Ez Config RM)

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5. Summary of test results

5.1 Summary of EMI emission test results

Applied	Test items	Test method	Result
☒	Conducted Emission	EN 55022:2010	Complied
☒	Radiated Emission	EN 55022:2010	Complied
☒	Harmonics current	EN 61000-3-2:2006+A2:2009	Complied
☒	Voltage fluctuations and flickers	EN 61000-3-3:2008	Complied

5.2 Summary of immunity test results

Applied	Test items	Test method	Result
*** EN 55024:2010**			
☒	Electrostatic discharge	EN 61000-4-2:2009	Complied
☒	Radiated RF immunity	EN 61000-4-3:2006+A2:2010	Complied
☒	Electric Fast Transient/BURST	EN 61000-4-4:2004+A1:2010	Complied
☒	Surge	EN 61000-4-5:2006	Complied
☒	Conducted RF immunity	EN 61000-4-6:2009	Complied
☐	Magnetic field immunity	EN 61000-4-8:2010	N/A
☒	Voltage dip/interruption	EN 61000-4-11:2004	Complied

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5.3 Performance criteria

Performance criterion A: The equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer when the equipment is used as intended.
The performance level may be replaced by a permissible loss of performance. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as intended.

Performance criterion B: After the test, the equipment shall continue to operate as intended without operator intervention. No degradation of performance or loss of function is allowed, after the application of the phenomena below a performance level specified by the manufacturer, when the apparatus is used as intended. The performance level may be replaced by a permissible loss of performance. During the test, degradation of performance is allowed. However, no change of operating state or stored data is allowed to persist after the test.
If the minimum performance level (or the permissible performance loss) is not specified by the manufacturer, then either of these may be derived from the product description and documentation, and by what the user may reasonably expect from the equipment if used as untended.

Performance criterion C: Loss of function is allowed, provided the function is self-recoverable or can be restored by the operating of the controls by the user In accordance with the manufacturer's instructions. Functions, and/or information stored in non-volatile memory, or protected by a battery backup, shall not be lost.

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6. Test results

6.1 Conducted Emission

Test specification	EN 55022:2010, Section 5, Class A		
Testing voltage	230 V, 50 Hz		
Test facility	Shielded room (CE#2)		
Date	2013. 08. 30		
Temperature (°C)	25 °C	Humidity (% R.H.)	42 % R.H.
Remarks	Complied		

6.1.1 Limits of conducted emission measurement

☒ AC main

Frequency [MHz]	Class A (dB(μV))		Class B (dB(μV))	
	Quasi-peak	Average	Quasi-peak	Average
0.15 ~ 0.5	79	66	66 ~ 56 *	56 ~ 46*
0.5 ~ 5	73	60	56	46
5 ~ 30	73	60	60	50

*The limit decreases linearly with the logarithm of frequency.

☐ Telecommunication

Frequency [MHz]	Class A Voltage Limits (dB(μV))		Current Limits (dB(μA))	
	Quasi-Peak	Average	Quasi-Peak	Average
0.15 ~ 0.5	97 to 87	84 to 74	53 to 43	40 to 30
0.5 ~ 30	87	74	43	30
Frequency [MHz]	Class B Limits (dB(μV))		Current Limits (dB(μA))	
	Quasi-Peak	Average	Quasi-Peak	Average
0.15 ~ 0.5	84 to 74	74 to 64	40 to 30	30 to 20
0.5 ~ 30	74	64	30	20

* The limits decrease linearly with the logarithm of the frequency in the range 0.15 MHz to 0.5 MHz

* The current and voltage disturbance limits are derived for use with an impedance stabilization Network (ISN) which presents a common mode (asymmetric mode) impedance of 150 Ω to the telecommunication port under test (conversion factor is 20 $\log_{10}$ 150/I = 44 dB).

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6.1.2 Measurement procedure

The measurements were performed in a shielded room. EUT was setup as shown in photograph and placed on a non-metallic table height of 0.8 m above the reference ground plane. The rear of table was located 0.4 m to the vertical conducted plane. EUT was power through the LISN, which was bonded to the ground plane. The LISN power was filtered. Each EUT power lead, except ground (safety) lead was individually connected through a LISN to input power source. EUT signal cables that hung closer than 0.4 m to the Horizontal metal ground 0.3 m ~ 0.4 m long. The power cord was bundles in the center. All peripheral equipment was powered from a sub LISN. The LISN and ISN were positioned 0.8 m from the EUT. Peak and Average detection were used in preliminary testing and Quasi-peak and Average detections were used at final measurement.
Both lines of power cord, hot and neutral, were measured.

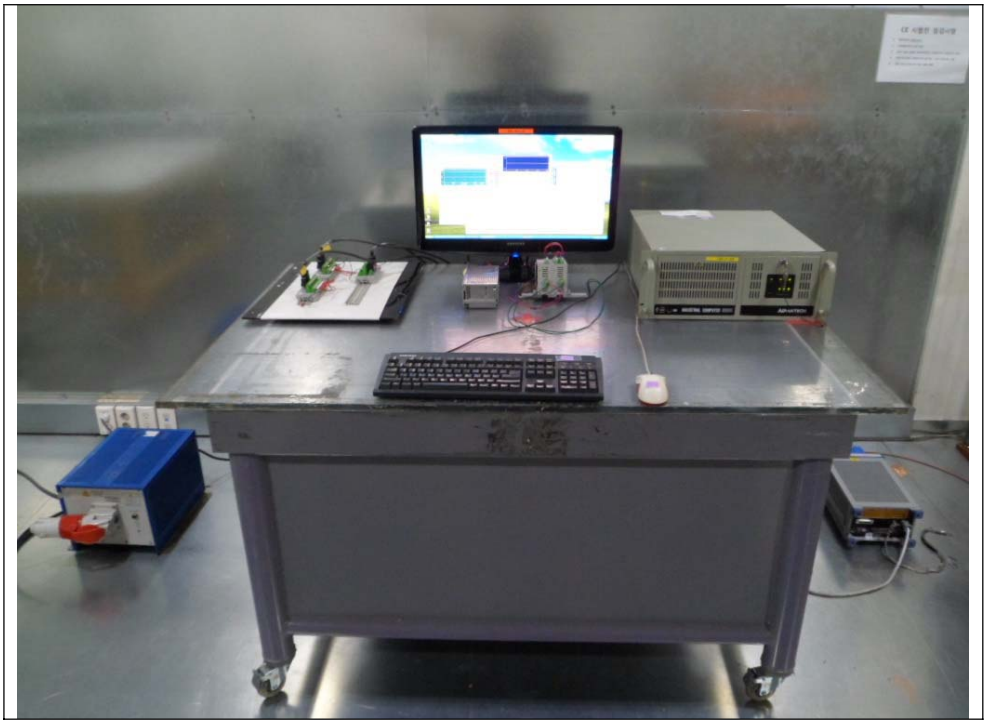
6.1.3 Used equipments

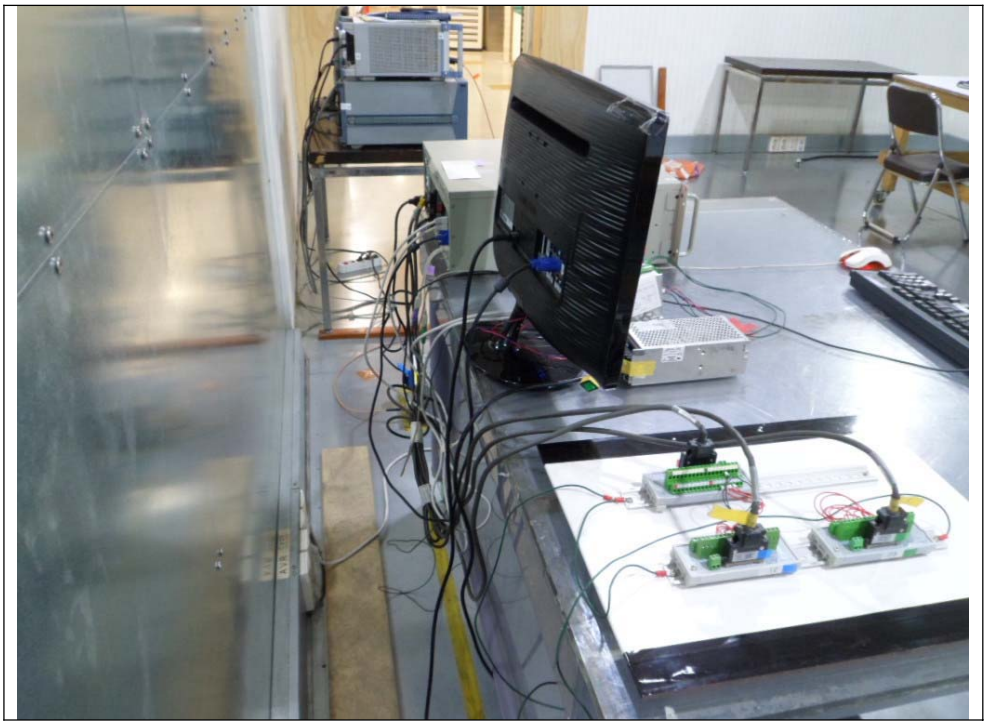
Equipment	Model	Serial No.	Makers	Next Cal. Date	Used
Test Receiver	ESHS30	844827/011	R&S	2014.08.05	☐
Test Receiver	ESCI7	100732	R&S	2014.02.18	☐
Test Receiver	ESCI	100001	R&S	2014.07.25	☐
Test Receiver	ESCI	100710	R&S	2013.11.06	☒
LISN	ENV216	101352	R&S	2014.01.07	☒
LISN	L3-32	0120J20305	PMM	-	☒
8-WIRE ISN	NTFM 8158 CAT5	CAT5-8158-0028	SCHWARZBECK	2014.04.13	☐
8-WIRE ISN	NTFM 8158 CAT3	CAT3-8158-0020	SCHWARZBECK	2014.04.13	☐
ISN	ST08	24342	TESEQ	2014.06.21	☐
ISN	ENY81	101545	R&S	2014.08.29	☐

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6.1.4 Photographs of test setup

* AC Main

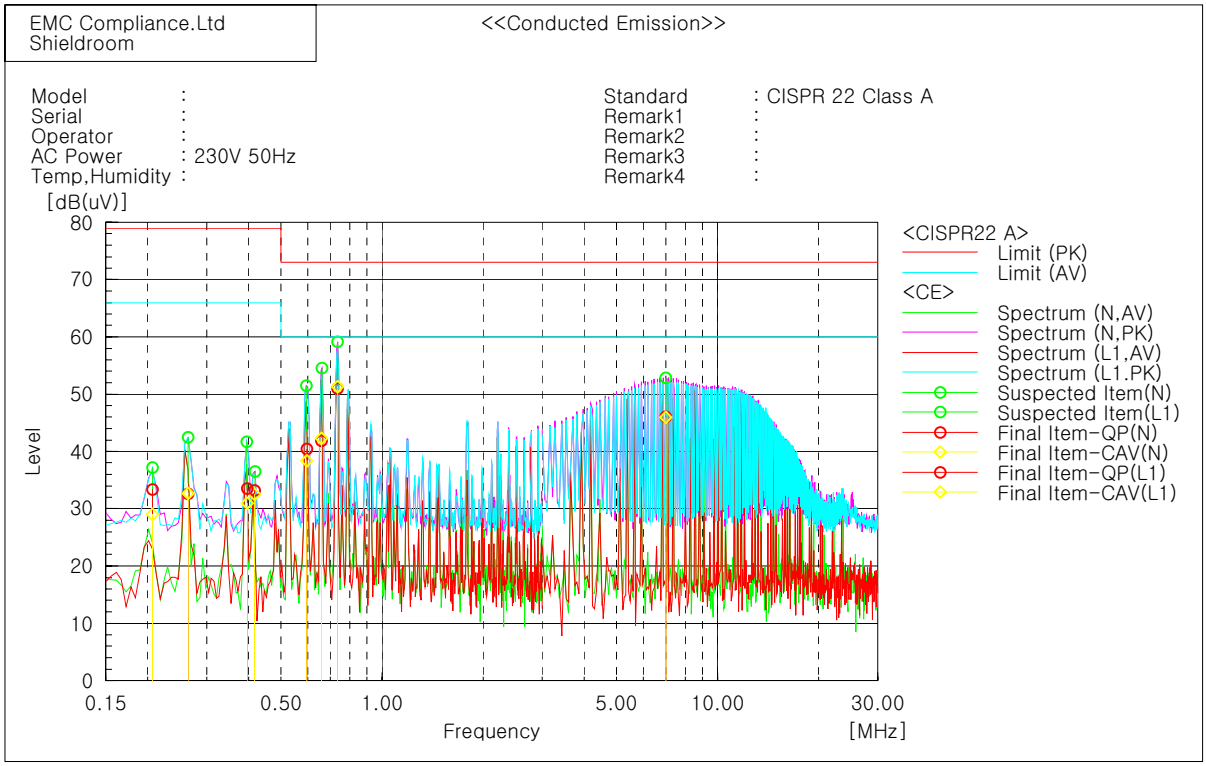




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6.1.5 Conducted emission measurement result

* AC Main (SIO-AI8F V1.0, SIO-AO4F V1.0, SIO-AI8AO4F V1.0)



Final Result

--- N Phase ---

No.	Frequency [MHz]	Reading QP [dB(uV)]	Reading CAV [dB(uV)]	c.f [dB]	Result QP [dB(uV)]	Result CAV [dB(uV)]	Limit QP [dB(uV)]	Limit AV [dB(uV)]	Margin QP [dB]	Margin CAV [dB]
1	0.41704	23.2	22.5	10.0	33.2	32.5	79.0	66.0	45.8	33.5
2	0.5957	30.6	28.6	9.8	40.4	38.4	73.0	60.0	32.6	21.6
3	0.66074	32.2	32.6	9.7	41.9	42.3	73.0	60.0	31.1	17.7
4	0.73626	41.3	41.6	9.7	51.0	51.3	73.0	60.0	22.0	8.7
5	7.00096	36.1	36.1	9.9	46.0	46.0	73.0	60.0	27.0	14.0

--- L1 Phase ---

No.	Frequency [MHz]	Reading QP [dB(uV)]	Reading CAV [dB(uV)]	c.f [dB]	Result QP [dB(uV)]	Result CAV [dB(uV)]	Limit QP [dB(uV)]	Limit AV [dB(uV)]	Margin QP [dB]	Margin CAV [dB]
1	0.20694	23.5	19.1	9.9	33.4	29.0	79.0	66.0	45.6	37.0
2	0.26404	22.7	22.8	9.9	32.6	32.7	79.0	66.0	46.4	33.3
3	0.39696	23.5	21.0	10.0	33.5	31.0	79.0	66.0	45.5	35.0

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6.2 Radiated Emission

Test specification	EN 55022:2010, Sections 6, Class A		
Testing voltage	230 V, 50 Hz		
Test facility	10 m Chamber (#F2)		
Test distance	10 m		
Date	2013. 08. 30		
Temperature (°C)	25 °C	Humidity (% R.H.)	50 % R.H.
Remarks	Complied		

6.2.1 Limits of radiated emission measurement

☒ Limits below 1 GHz

Frequency [MHz]	Class A (dB(μV/m)) @ 10 m	Class B (dB(μV/m)) @ 10 m
30 ~ 230	40	30
230 ~ 1 000	47	37

☐ Limits above 1 GHz

Frequency [MHz]	Class A @ 3 m		Class B @ 3 m	
	Average limit (dB(μV/m))	Peak limit (dB(μV/m))	Average limit (dB(μV/m))	Peak limit (dB(μV/m))
1 ~ 3	56	76	50	70
3 ~ 6	60	80	54	74
Note - The lower limit applies at the transition frequency.				

6.2.2 Measurement procedure

The test was done at a 10 m chamber with a quasi-peak detector. EUT was placed on a non-metallic table height of 0.8 m above the reference ground plane. Cables were folded back and forth forming a bundle 0.3 m to 0.4 m long and were hanged at a 0.4 m height to the ground plane.
Cables connected to EUT were fixed to cause maximum emission. Test was made with the antenna positioned in both the horizontal and vertical planes of polarization. The measurement antenna was varied in height above the conducting ground plane to obtain the maximum signal strength.

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6.2.3 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. Date	Used
Test Receiver	ESCI7	100732	R&S	2014.02.18	☒
Test Receiver	ESCI	100001	R&S	2014.07.25	☐
Test Receiver	ESCI	100710	R&S	2013.11.06	☐
Bi-Log Antenna	VULB 9168	440	SCHWARZBECK	2013.10.04	☒
Amplifier	310N	293004	SONOMA INSTR UMENT	2013.11.06	☒
3 dB Attenuator	8491B	22981	HP	2014.03.19	☒
Antenna Mast	MA4000-EP	303	Innco Systems	-	☒
Turn Table	DT2000S-1t	079	Innco Systems	-	☒
Amplifier	8449B	3008A02343	AGILENT	2013.11.06	☐
Horn ANT	3115	00086706	ETS	2013.11.21	☐
Spectrum Analyzer	FSP7	100289	R&S	2013.12.14	☐

6.2.4 Sample calculation

The field strength is calculated adding the antenna Factor, cable loss and, Antenna pad adding, subtracting the amplifier gain from the measured reading.

The sample calculation is as follow:

Result = M.R + C.F(A.F + C.L +3 dB Att – A.G)

M.R = Meter Reading

C.F = Correction Factor

A.F = Antenna Factor

C.L = Cable Loss

A.G= Amplifier Gain

3 dB Att = 3 dB Attenuator

If M.R is 30 dB, A.F 12 dB, C.L 5 dB, 3 dB, A.G 35 dB

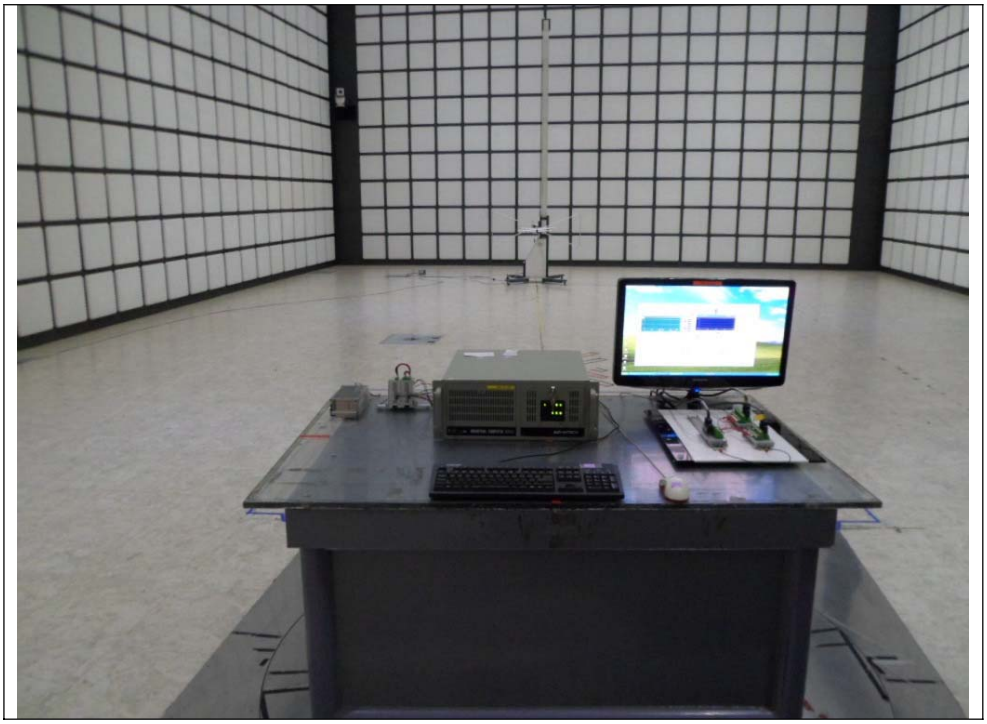
The result is

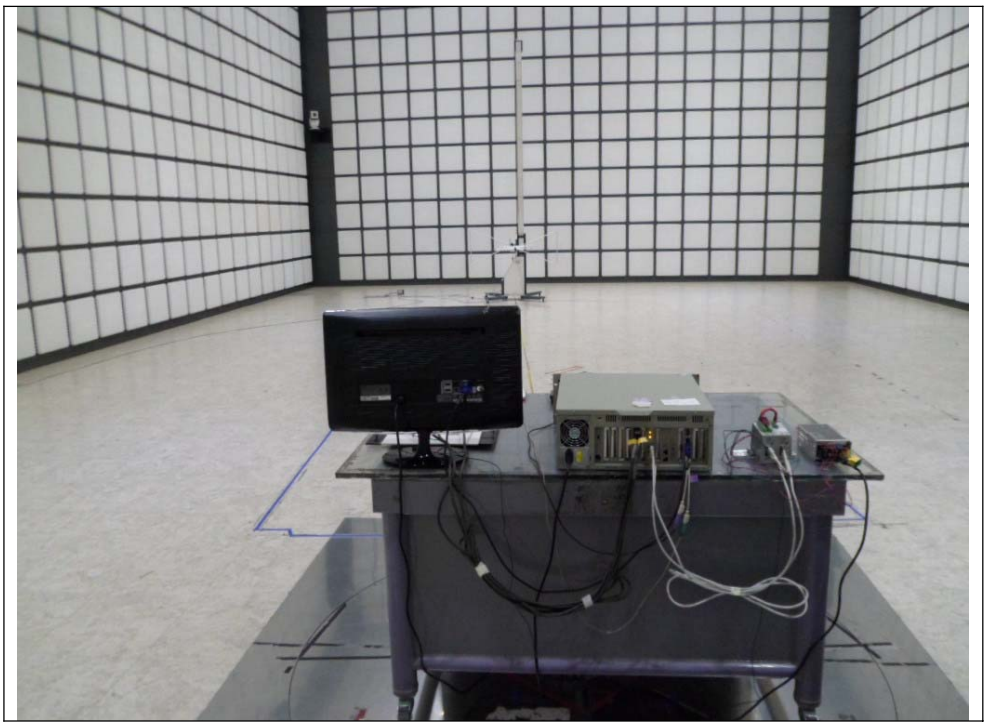
30 + 12 + 5 + 3 – 35 = 15 dB(μV/m)

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6.2.5 Photographs of test setup

* 30 ㎒ ~ 1 ㎓



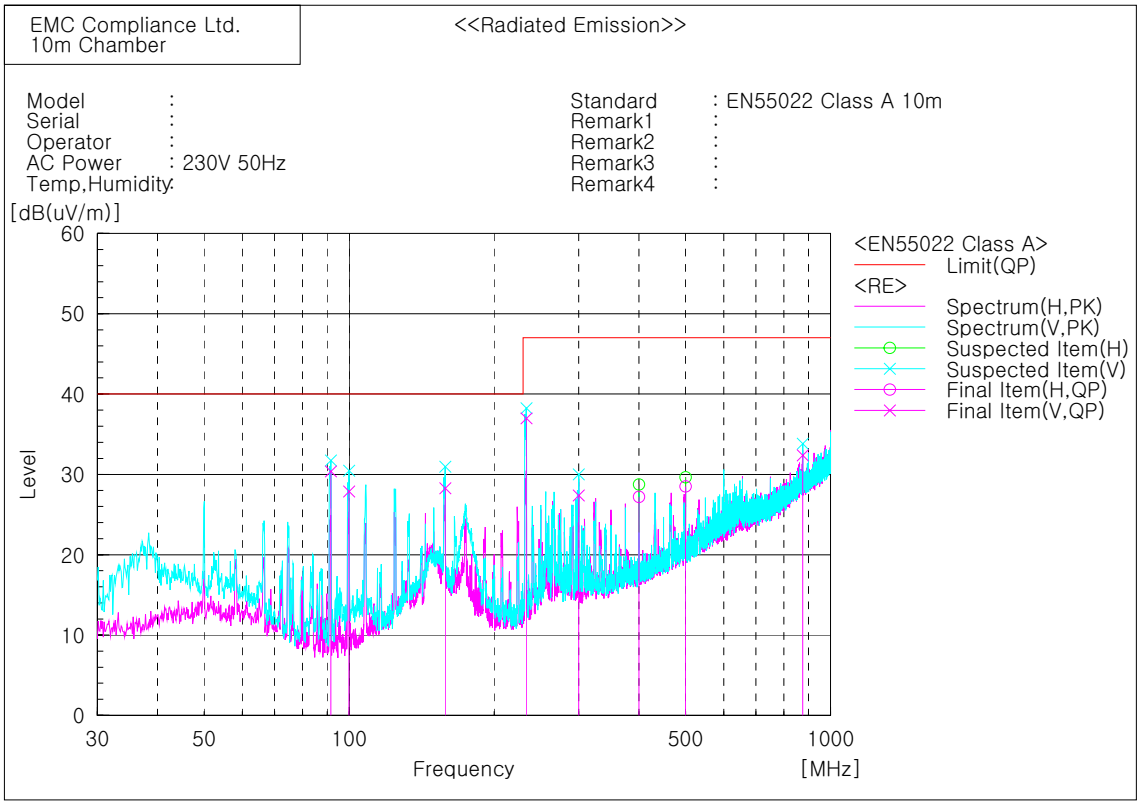


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6.2.6 Radiated emission measurement result

*** Graph and Data**

* 30 ㎒ ~ 1 ㎓ (SIO-AI8F V1.0, SIO-AO4F V1.0, SIO-AI8AO4F V1.0)



Final Result

No.	Frequency [MHz]	(P)	Reading QP [dB(uV)]	c.f [dB(1/m)]	Result QP [dB(uV/m)]	Limit QP [dB(uV/m)]	Margin QP [dB]	Height [cm]	Angle [deg]
1	91.595	V	48.8	-18.4	30.4	40.0	9.6	199.0	144.6
2	99.961	V	45.4	-17.5	27.9	40.0	12.1	100.0	175.4
3	158.283	V	40.5	-12.2	28.3	40.0	11.7	100.0	22.1
4	233.336	V	50.4	-13.4	37.0	47.0	10.0	100.0	195.5
5	300.024	V	37.8	-10.4	27.4	47.0	19.6	100.0	130.1
6	400.055	H	34.7	-7.5	27.2	47.0	19.8	201.0	288.2
7	499.965	H	33.0	-4.5	28.5	47.0	18.5	201.0	323.4
8	875.112	V	28.7	3.7	32.4	47.0	14.6	199.0	345.9

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6.3 Harmonics

Test specification	EN 61000-3-2:2006+A2:2009				
Testing voltage	230 V, 50 Hz				
Test facility	Immunity area				
Date	2013. 09. 02				
Temperature(°C)	25 °C	Humidity (% R.H.)	38 % R.H.	Pressure (kPa)	99.7 kPa
Remarks	Complied				

6.3.1 Measurement procedure

The equipment is supplied in series with shunt(s) Rm or current transformer(s) from a source having the same nominal voltage and frequency as the rated supply voltage and frequency of the equipment. Measurements shall be made under normal load, or conditions for adequate heat discharge, and under normal operating conditions. User's operation controls or automatic programmers shall be set to produce the maximum harmonic component, for each successive harmonic component in turn. For the purpose of harmonic current limitation, equipment is classified as follows :

Class A : Equipment not specified in one of the three other Classes shall be considered as Class A equipment.

- Balanced three-phase equipment;
- Household appliances excluding equipment identified as Class D;
- Tools excluding portable tools;
- Dimmers for incandescent lamps;
- Audio equipment.

Class B : Portable tools; Arc welding equipment which is not professional equipment.

Class C : Lighting equipment.

Class D : Equipment having a specified power according to 6.2.2 less than or equal to 600 w, of the following types:

- Personal computers and personal computer monitors;
- Television receivers.

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6.3.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Harmonics/Flicker meter	5001x-CTS -400-413	54984	C.I.	2014.05.22	☒

6.3.3 Photographs of test setup



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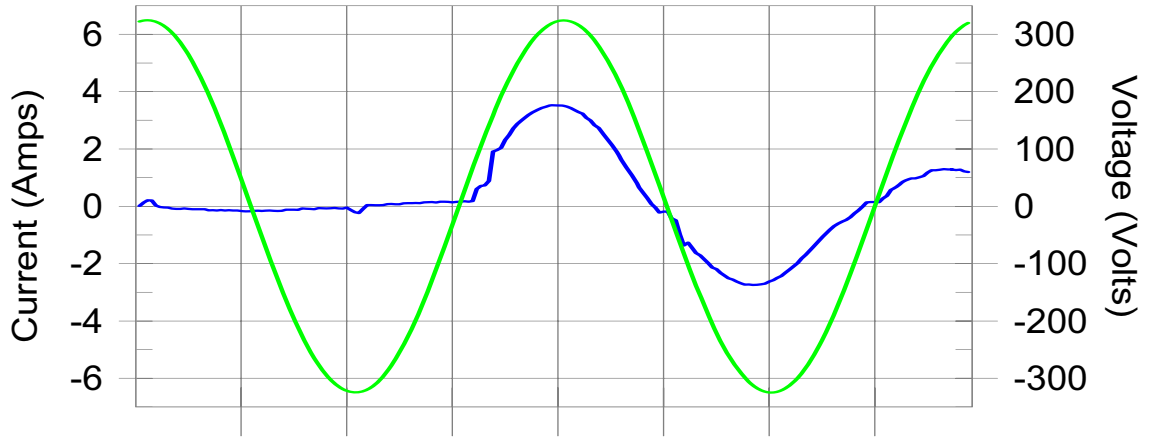
6.3.4 Measurement result

Harmonics – Class-A per Ed. 3.2 (2009)(Run time) incl. inter-harmonics

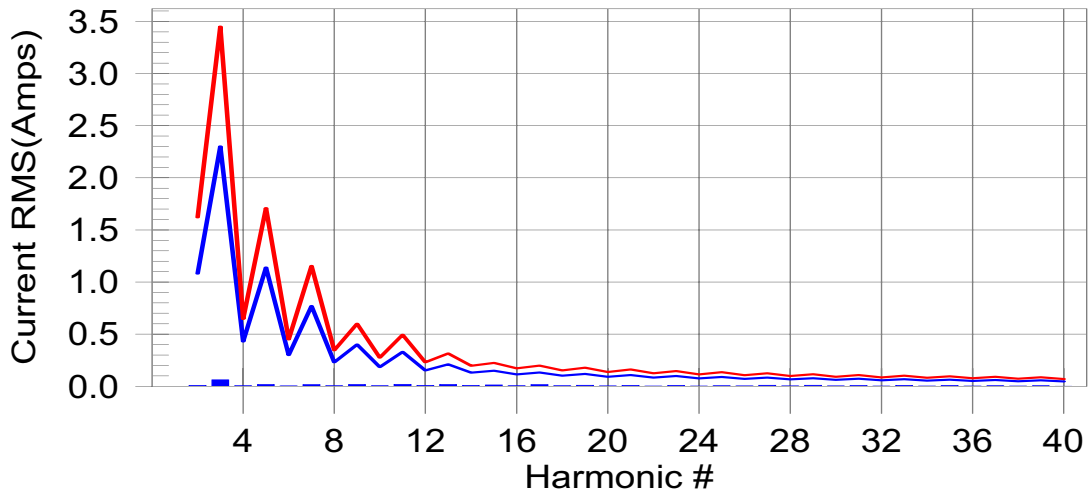
EUT: SIO-AI8F V1.0, SIO-AO4F V1.0, SIO-AI8AO4F V1.0 **Tested by: Test Operator**
Test category: Class-A per Ed. 3.2 (2009) (European limits) **Test Margin: 100**
Test date: 2013-09-02 **Start time: 오전 7:35:47** **End time: 오전 7:38:37**
Test duration (min): 2.5 **Data file name: H-000178.cts_data**
Comment: Comments
Customer: AJINEXTEK CO.,LTD

Test Result: Pass **Source qualification: Normal**

Current & voltage waveforms



Harmonics and Class A limit line **European Limits**



Test result: Pass **Worst harmonic was #29 with 8.93% of the limit.**

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Current Test Result Summary (Run time)

EUT: SIO-AI8F V1.0, SIO-AO4F V1.0, SIO-AI8AO4F V1.0 **Tested by: Test Operator**
Test category: Class-A per Ed. 3.2 (2009) (European limits) **Test Margin: 100**
Test date: 2013-09-02 **Start time: 오전 7:35:47** **End time: 오전 7:38:37**
Test duration (min): 2.5 **Data file name: H-000178.cts_data**
Comment: Comments
Customer: AJINEXTEK CO.,LTD

Test Result: Pass **Source qualification: Normal**
THC(A): 0.06 **I-THD(%): 13.06** **POHC(A): 0.007** **POHC Limit(A): 0.305**
Highest parameter values during test:

V_RMS (Volts):	229.68	Frequency(Hz):	50.00
I_Peak (Amps):	3.549	I_RMS (Amps):	0.724
I_Fund (Amps):	0.569	Crest Factor:	4.901
Power (Watts):	123.8	Power Factor:	0.943

Harm#	Harms(avg)	100%Limit	%of Limit	Harms(max)	150%Limit	%of Limit	Status
2	0.003	1.080	0.3	0.009	1.620	0.58	Pass
3	0.051	2.300	2.2	0.062	3.450	1.79	Pass
4	0.002	0.430	0.0	0.004	0.645	0.70	Pass
5	0.013	1.140	1.1	0.019	1.710	1.09	Pass
6	0.001	0.300	0.0	0.004	0.450	0.83	Pass
7	0.013	0.770	1.7	0.016	1.155	1.42	Pass
8	0.002	0.230	0.9	0.005	0.345	1.49	Pass
9	0.011	0.400	2.6	0.017	0.600	2.85	Pass
10	0.002	0.184	1.3	0.006	0.276	2.23	Pass
11	0.015	0.330	4.4	0.019	0.495	3.94	Pass
12	0.003	0.153	1.9	0.007	0.230	3.23	Pass
13	0.011	0.210	5.0	0.018	0.315	5.64	Pass
14	0.003	0.131	2.4	0.008	0.197	4.24	Pass
15	0.008	0.150	5.6	0.013	0.225	5.89	Pass
16	0.002	0.115	1.9	0.006	0.173	3.20	Pass
17	0.011	0.132	8.3	0.016	0.199	8.21	Pass
18	0.002	0.102	0.0	0.005	0.153	3.14	Pass
19	0.006	0.118	4.8	0.010	0.178	5.37	Pass
20	0.001	0.092	0.0	0.004	0.138	2.77	Pass
21	0.003	0.107	7.7	0.008	0.161	5.27	Pass
22	0.001	0.084	0.0	0.003	0.125	2.07	Pass
23	0.005	0.098	5.4	0.007	0.147	5.06	Pass
24	0.001	0.077	0.0	0.003	0.115	2.68	Pass
25	0.003	0.090	3.6	0.006	0.135	4.33	Pass
26	0.002	0.071	0.0	0.004	0.106	3.31	Pass
27	0.005	0.083	5.8	0.010	0.125	7.84	Pass
28	0.002	0.066	0.0	0.004	0.099	4.17	Pass
29	0.006	0.078	7.8	0.010	0.116	8.93	Pass
30	0.002	0.061	0.0	0.004	0.092	4.40	Pass
31	0.004	0.073	5.8	0.007	0.109	6.19	Pass
32	0.002	0.058	0.0	0.003	0.086	3.40	Pass
33	0.004	0.068	6.0	0.008	0.102	7.64	Pass
34	0.001	0.054	0.0	0.003	0.081	3.16	Pass
35	0.004	0.064	6.4	0.007	0.096	6.85	Pass
36	0.001	0.051	0.0	0.003	0.077	3.28	Pass
37	0.003	0.061	0.0	0.005	0.091	5.38	Pass
38	0.001	0.048	0.0	0.002	0.073	3.12	Pass
39	0.003	0.058	5.9	0.006	0.087	6.34	Pass
40	0.001	0.046	0.0	0.002	0.069	2.55	Pass

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Voltage Source Verification Data (Run time)

EUT: SIO-AI8F V1.0, SIO-AO4F V1.0, SIO-AI8AO4F V1.0 **Tested by: Test Operator**
Test category: Class-A per Ed. 3.2 (2009) (European limits) **Test Margin: 100**
Test date: 2013-09-02 **Start time: 오전 7:35:47** **End time: 오전 7:38:37**
Test duration (min): 2.5 **Data file name: H-000178.cts_data**
Comment: Comments
Customer: AJINEXTEK CO.,LTD

Test Result: Pass **Source qualification: Normal**

Highest parameter values during test:

Voltage (Vrms):	229.68	Frequency(Hz):	50.00
I_Peak (Amps):	3.549	I_RMS (Amps):	0.724
I_Fund (Amps):	0.569	Crest Factor:	4.901
Power (Watts):	123.8	Power Factor:	0.943

Harm#	Harmonics V-rms	Limit V-rms	% of Limit	Status
2	0.099	0.459	21.50	OK
3	0.565	2.067	27.35	OK
4	0.021	0.459	4.67	OK
5	0.023	0.919	2.47	OK
6	0.026	0.459	5.76	OK
7	0.024	0.689	3.46	OK
8	0.018	0.459	3.96	OK
9	0.019	0.459	4.12	OK
10	0.010	0.459	2.23	OK
11	0.027	0.230	11.77	OK
12	0.022	0.230	9.60	OK
13	0.028	0.230	12.21	OK
14	0.010	0.230	4.51	OK
15	0.015	0.230	6.54	OK
16	0.012	0.230	5.29	OK
17	0.020	0.230	8.54	OK
18	0.017	0.230	7.56	OK
19	0.009	0.230	4.03	OK
20	0.005	0.230	2.09	OK
21	0.012	0.230	5.41	OK
22	0.009	0.230	3.78	OK
23	0.010	0.230	4.18	OK
24	0.006	0.230	2.81	OK
25	0.018	0.230	7.81	OK
26	0.010	0.230	4.14	OK
27	0.018	0.230	7.94	OK
28	0.010	0.230	4.23	OK
29	0.016	0.230	6.86	OK
30	0.008	0.230	3.68	OK
31	0.015	0.230	6.57	OK
32	0.009	0.230	3.92	OK
33	0.013	0.230	5.50	OK
34	0.010	0.230	4.51	OK
35	0.014	0.230	6.05	OK
36	0.007	0.230	3.25	OK
37	0.013	0.230	5.51	OK
38	0.009	0.230	3.77	OK
39	0.018	0.230	7.73	OK
40	0.010	0.230	4.43	OK

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6.4 Flicker

Test specification	EN 61000-3-3:2008				
Testing voltage	230 V, 50 Hz				
Test facility	Immunity area				
Date	2013. 09. 02				
Temperature(°C)	25 °C	Humidity (% R.H.)	38 % R.H.	Pressure (kPa)	99.7 kPa
Remarks	Complied				

6.4.1 Measurement procedure

EUT was connected to the power analyzer system.

Measurement was performed to obtain the desired flicker parameters.

The measuring time depends on which parameters are to be measured.

P_{lt} = 2 h

P_{st} = 10 min

Controls and automatic programs shall be set to produce the most unfavorable sequence of voltage changes, using only those combinations of controls and programs are mentioned by the manufacturer in the instruction manual.

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6.4.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Harmonics/Flicker meter	5001x-CTS -400-413	54984	C.I.	2014.05.22	☒

6.4.3 Photographs of test setup



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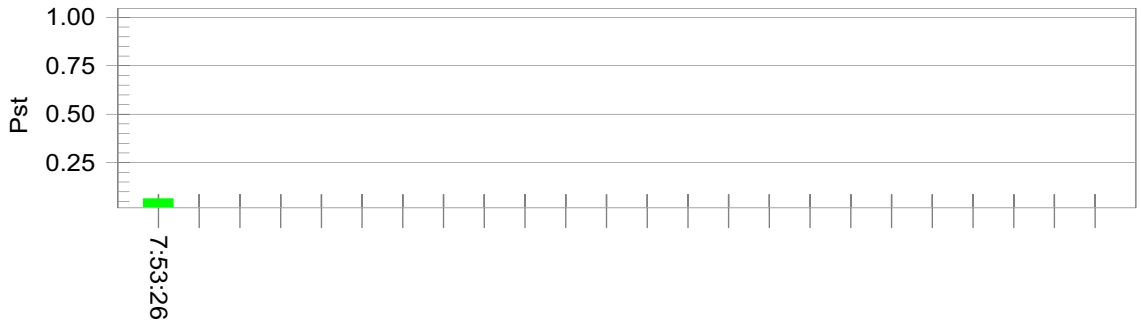
6.4.4 Measurement result

Flicker Test Summary per EN/IEC61000-3-3 (Run time)

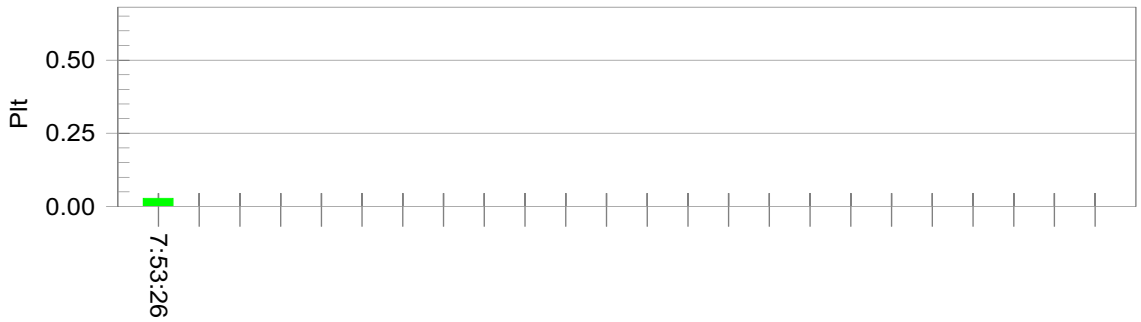
EUT: SIO-AI8F V1.0, SIO-AO4F V1.0, SIO-AI8AO4F V1.0 **Tested by: Test Operator**
Test category: All parameters (European limits) **Test Margin: 100**
Test date: 2013-09-02 **Start time: 오전 7:43:06** **End time: 오전 7:53:27**
Test duration (min): 10 **Data file name: F-000179.cts_data**
Comment: Comments
Customer: AJINEXTEK CO.,LTD

Test Result: Pass **Status: Test Completed**

Pst_i and limit line **European Limits**



Plt and limit line



Parameter values recorded during the test:

Vrms at the end of test (Volt):	**229.66**			
Highest dt (%):	**0.00**	**Test limit (%):**	**3.30**	**Pass**
Time(mS) > dt:	**0.0**	**Test limit (mS):**	**500.0**	**Pass**
Highest dc (%):	**0.00**	**Test limit (%):**	**3.30**	**Pass**
Highest dmax (%):	**0.00**	**Test limit (%):**	**4.00**	**Pass**
Highest Pst (10 min. period):	**0.064**	**Test limit:**	**1.000**	**Pass**
Highest Plt (2 hr. period):	**0.028**	**Test limit:**	**0.650**	**Pass**

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6.5 Electrostatic Discharge

Test specification	EN 61000-4-2:2009, Criteria : B				
Test level	☒ Contact: ± 4 kV ☒ Air: ± 2 kV, ± 4 kV, ± 8 kV ☒ HCP: ± 2 kV, ± 4 kV ☒ VCP: ± 2 kV, ± 4 kV				
Discharge impedance	330 Ω / 150 pF				
Number of discharge (Each polarity)	☒ Contact: 25 ☒ Air: 10 ☒ HCP / VCP: 25				
Interval between discharges	1 s				
Testing voltage	230 V, 50 Hz				
Test facility	Shielded room				
Date	2013. 09. 02				
Temperature(°C)	25 °C	Humidity (% R.H.)	45 % R.H.	Pressure (kPa)	99.7 kPa
Remarks	Complied - A: There was no change of operation status during above testing.				

6.5.1 Measurement procedure

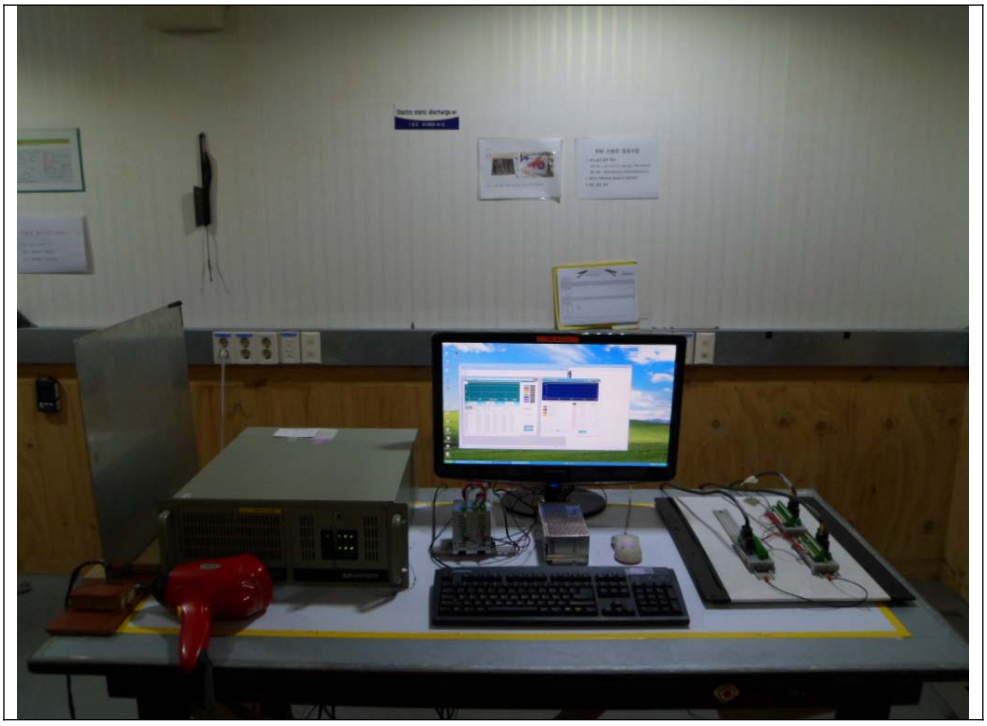
A ground reference plane was located on the floor, and connected to earth via a low Impedance connection. The return cable of the ESD generator was connected to the reference plane.
In case of floor standing equipment, EUT was placed on the reference plane on 0.1 m of insulating Support. In case of table top equipment, EUT was placed on a wooden table 0.8 m above the reference grounded floor. A horizontal coupling plane (HCP) was placed on the table, and Connected to the reference plane via a 470 kΩ resistor located in each end (0.5 mm insulating support between EUT and HCP). In both cases a vertical coupling plane(VCP) OF 0.5 X 0.5 m was located 0.1 m from the EUT's sides. The VCP was connected to the reference plane in the same matter as the HCP.

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6.5.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. Date	Used
ESD Tester	PESD 1600	H011 309	HAEFELY	2014.07.15	☐
ESD Tester	NSG 437	182	TESEQ	2014.05.21	☒
HCP	-	-	-	-	☒
VCP	-	-	-	-	☒

6.5.3 Photographs of test setup

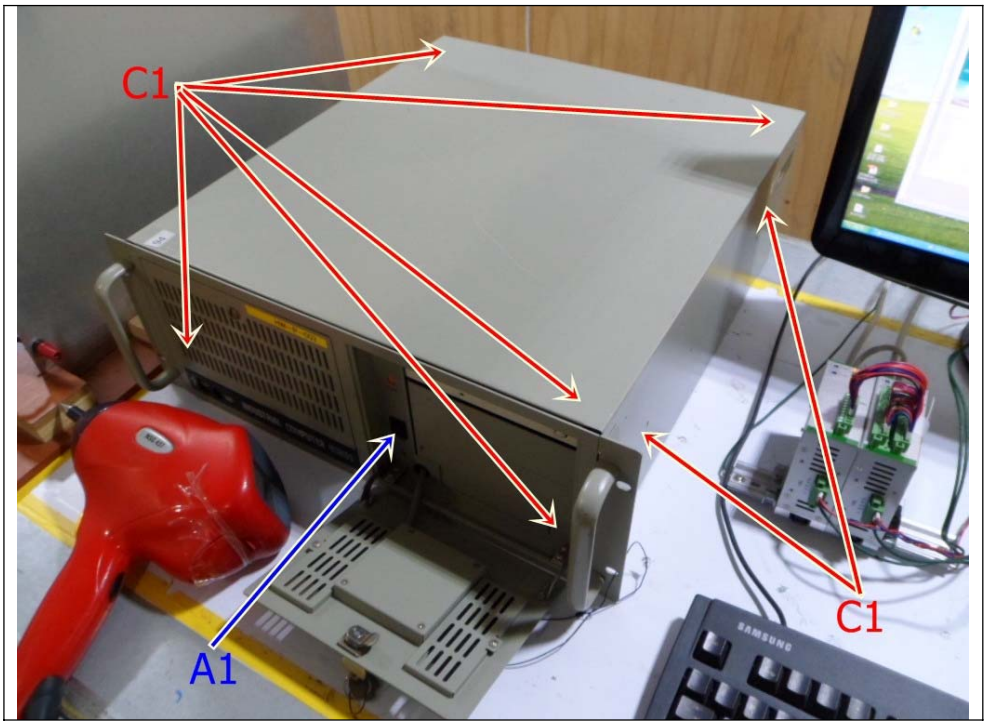


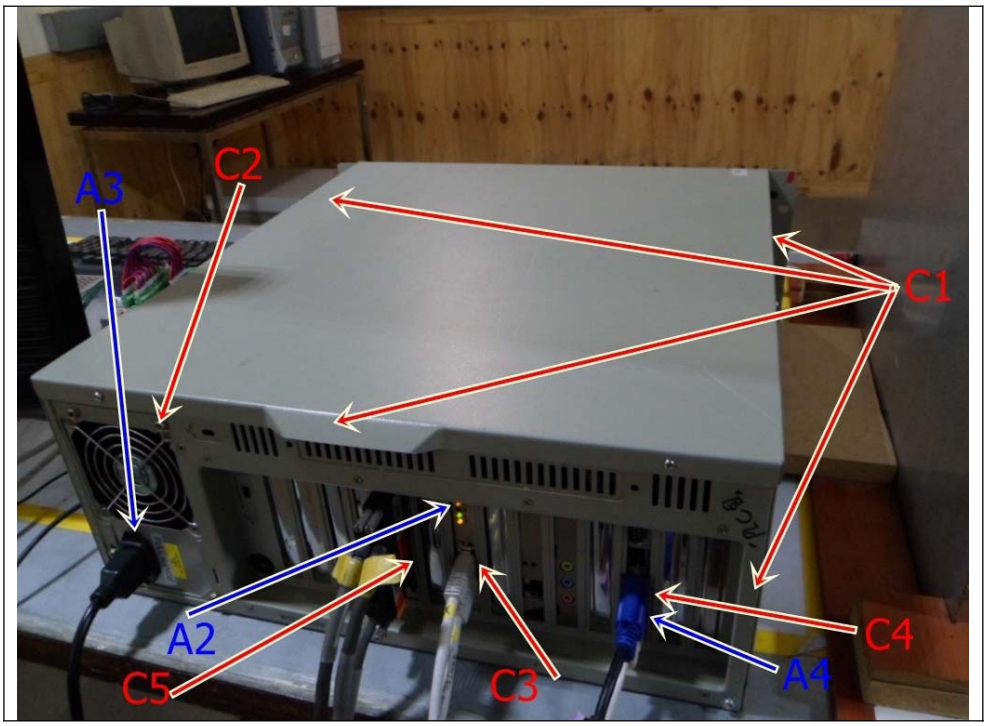
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6.5.4 Measurement result

Electrostatic Discharge (Test Point)

Air discharge →
Contact discharge →





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HCP/VCP discharge

Location(EUT)	Applied level (±)	Result (Criterion)
HCP (All 4 sides)	± 2 kV, ± 4 kV	A
VCP (All 4 sides)	± 2 kV, ± 4 kV	A

Contact discharge

Location(EUT)		Applied level (±)	Result (Criterion)
C1	Enclosure(Case)	± 4 kV	A
C2	Screw	± 4 kV	A
C3	TX/RX Port	± 4 kV	A
C4	D-SUB Port	± 4 kV	A

Air discharge

Location(EUT)		Applied level (±)	Result (Criterion)
A1	Switch	± 2 kV, ± 4 kV, ± 8 kV	A
A2	LED	± 2 kV, ± 4 kV, ± 8 kV	A
A3	Power Port	± 2 kV, ± 4 kV, ± 8 kV	A
A4	PS/2 Port	± 2 kV, ± 4 kV, ± 8 kV	A

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6.6 Radio Frequency Electromagnetic Fields

Test specification	EN 61000-4-3:2006+A2:2010, Criteria : A				
Tested frequency	80 ㎒ ~ 1 ㎓				
Test level & Modulation	3 V/m, 80 % Amplitude Modulation (1 ㎑)				
Frequency Step	log 1 % step				
Dwell time	3 s				
Distance	3 m from EUT to tip of antenna				
Testing Voltage	230 V, 50 ㎐				
Test facility	Fully anechoic chamber (3 m)				
Date	2013. 08. 30				
Temperature(°C)	25 °C	Humidity (% R.H.)	38 % R.H.	Pressure (㎪)	99.7 ㎪
Remarks	Complied - A: There was no change of operation status during above testing.				

6.6.1 Measurement procedure

The test was performed at 3 m full anechoic chamber.

For floor standing equipment, the EUT was standing on the floor.

For tabletop equipment, the EUT was located on a wooden table 0.8 m above the floor.

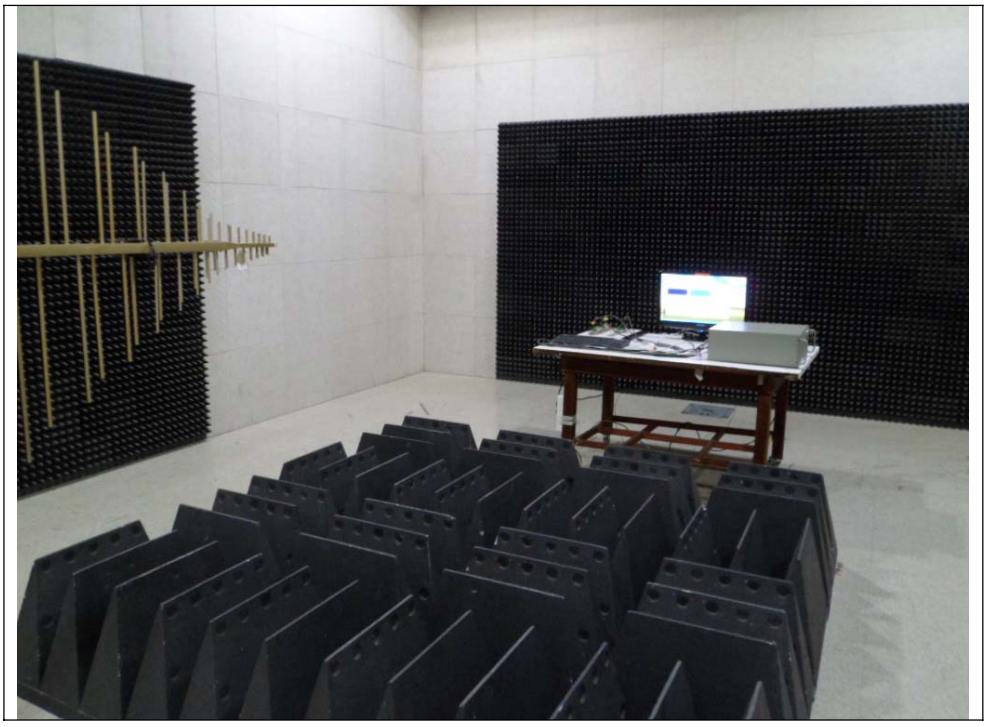
The EUT was tested all sides, horizontal and vertical polarization.

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6.6.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Power meter	PM2002	302852	AR	2014.04.05	☒
Power sensor	PH2000	303224	AR	2014.04.05	☒
Power sensor	PH2000	311217	AR	2014.04.05	☒
Directional coupler	DC6180	303976	AR	2014.04.08	☒
Signal generator	E4421B	GB40052295	AGILENT	2013.10.11	☒
Amplifier	BBA100	100996-1	R&S	2014.02.12	☒
Broadband Ant.	LPDA-0803	130269	ETS	-	☒
Fiber Optic Modem	HI-4413P	-	ETS-LINDG REM	-	☒
Antenna master	-	-	ETS	-	☒

6.6.3 Photographs of test setup



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6.6.4 Measurement result

Location(EUT)	Antenna polarization	Result (Criterion)
Front side	Horizontal	A
	Vertical	A
Rear side	Horizontal	A
	Vertical	A
Left side	Horizontal	A
	Vertical	A
Right side	Horizontal	A
	Vertical	A

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6.7 Electric Fast Transient/BURST

Test specification	EN 61000-4-4:2004+A1:2010, Criteria : B				
Coupling	☒ AC main ☒ Signal: Clamp ☐ Telecommunication: Clamp				
Test level	☒ AC main: ± 1 kV Peak ☒ Signal: ± 0.5 kV Peak ☐ Telecommunication: ± 0.5 kV Peak				
Repetition frequency	5 kHz, Tr/Th = 5 / 50 ns				
Coupling time	60 s				
Testing Voltage	230 V, 50 Hz				
Test facility	Shielded room				
Date	2013. 08. 31				
Temperature(°C)	27 °C	Humidity (% R.H.)	48 % R.H.	Pressure (kPa)	100.8 kPa
Remarks	Complied - A: There was no change of operation status during above testing.				

6.7.1 Measurement procedure

A ground reference plane was located on the floor.
EFT generator was connected to reference ground plane via low impedance connection.
For floor standing equipment, EUT was placed on a 0.1 m wooden table.
For tabletop equipment, EUT was placed on a 0.1 m above the ground reference plane.
Test generator and coupling/decoupling network was placed on, and bounded to, the ground reference plane. When using the coupling clamp, the minimum distance between the coupling plates and all other conductive surfaces, except the ground reference plane beneath the coupling clamp, Shall be 0.5 m.

6.7.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. date	Used
Ultra compact simulator	UCS 500-M	V0545100858	EM TEST	2014.01.31	☐
Ultra compact simulator	UCS500M	0701-03	EM TEST	2014.06.21	☒
Capacitive Coupling Clamp	-	-	EM TEST	-	☒

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6.7.3 Photographs of test setup





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6.7.4 Measurement result

* AC main

EFT coupling point	(+)	(-)	Result (Criterion)
L+N+PE	+ 1 kV	- 1 kV	A

* Signal

EFT coupling point	(+)	(-)	Result (Criterion)
PCI	+ 0.5 kV	- 0.5 kV	A

* Telecommunication

EFT coupling point	(+)	(-)	Result (Criterion)
-	-	-	-

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6.8 Surge

Test specification	EN 61000-4-5:2006, Criteria : B				
Coupling	☒ AC main: Direct ☐ Signal: Direct / CDN				
Test level	☒ AC main: ☒Differential mode: ± 0.5 kV, ± 1 kV ☒Common mode: ± 0.5 kV, ± 1 kV, ± 2 kV ☐ Signal: ± 0.5 kV, ± 1 kV				
Coupling Impedance	☒ Differential mode: 18 μF ☒ Common mode: 10 Ω + 9 μF ☐ 40 Ω + 0.5 μF ☐ Direct				
Surge pulse shape	Tr/Th = 1.2 / 50 μs				
Angles	0 °, 90 °, 180 °, 270 °				
Number of surge	5				
Coupling time	1 min				
Testing Voltage	230 V, 50 Hz				
Test facility	Shielded room				
Date	2013. 08. 31				
Temperature(°C)	27 °C	Humidity (% R.H.)	48 % R.H.	Pressure (kPa)	100.8 kPa
Remarks	Complied - A: There was no change of operation status during above testing.				

6.8.1 Measurement procedure

A ground reference plane was located on the floor. SURGE generator was connected to reference ground plane via low impedance connection. For floor standing equipment & table top equipment, EUT was placed on a wooden table.

6.8.2 Used equipments

Equipment	Model No.	Serial No.	Makers	Next Cal. date	Used
Ultra compact simulator	UCS 500-M	V0545100858	EM TEST	2014.01.31	☐
Ultra compact simulator	UCS500M	0701-03	EM TEST	2014.06.21	☒
CDN	CNV 508 N1	V1108108861	EM TEST	2014.01.03	☐

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6.8.3 Photographs of test setup



6.8.4 Measurement result

* AC main

Coupling point	(+)	(-)	Result (Criterion)
L+N	+ 0.5 kV, + 1 kV	- 0.5 kV, - 1 kV	A
L+PE	+ 0.5 kV, + 1 kV, + 2 kV	- 0.5 kV, - 1 kV, - 2 kV	A
N+PE	+ 0.5 kV, + 1 kV, + 2 kV	- 0.5 kV, - 1 kV, - 2 kV	A

* Signal

Coupling point	(+)	(-)	Result (Criterion)
-	-	-	-

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6.9 Conducted Immunity

Test specification	EN 61000-4-6:2009, Criteria : A				
Tested frequency	0.15 ㎒ ~ 80 ㎒				
Test level & Modulation	3 Ⅴ, 80 % Amplitude Modulation (1 ㎑)				
Frequency Step	log 1 % step				
Coupling method	☒ AC main: CDN(M3) ☒ Signal: Clamp ☐ Telecommunication: Clamp				
Testing Voltage	230 Ⅴ, 50 ㎐				
Test facility	Shielded room				
Date	2013. 08. 31				
Temperature(°C)	25 °C	Humidity (% R.H)	45 % R.H	Pressure(㎪)	99.7 ㎪
Remarks	Complied - A: There was no change of operation status during above testing.				

6.9.1 Measurement procedure

A ground reference plane was located on the floor.

The test was performed on a ground reference plane on a 0.1 m wooden table. This test were Performed using CDN for mains, clamp for signal and injection probe. The frequency range was swept from 0.15 ㎒ to 80 ㎒. This frequency range was Modulated with 1 ㎑ sine wave at 80 %.

The signal generators provided the modulated frequency at a 1 % step size.

The power and all network cable, I/O cables longer than 3 m length were tested.

6.9.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
CS generator	CWS 500	V0635101750	EM TEST	2014.01.17	☒
CDN	CDN L-801 M2/M3	2936	EM TEST	2014.02.06	☒
CDN	CDN M2/M3	0906-12	EM TEST	2013.10.11	☒
Attenuator	73-6-34	MU918	MCE/WEINSCHEL	2013.10.11	☒
EM Clamp	KEMZ 801	17643	Schaffner	2014.04.17	☒
CDN	CDN S1/75	0410-28	EM TEST	2014.05.13	☐
CDN	CDN-T8-RJ45	0113-22	EM TEST	2014.02.22	☐

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6.9.3 Photographs of test setup





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6.9.4 Measurement result

* AC main

Coupling point	Coupling method	Result (Criterion)
Power	CDN(M3)	A

* Signal

Coupling point	Coupling method	Result (Criterion)
PCI	Clamp	A

* Telecommunication

Coupling point	Coupling method	Result (Criterion)
-	-	-

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6.10 Dips and Interruptions

Test specification	EN 61000-4-11:2004 , Criterion: B or C				
Number of dips	3 T				
Duration	10 s				
Phase	Zero crossing (0 °, 180 °)				
Testing Voltage	230 V, 50 Hz				
Test facility	Shielded room				
Date	2013. 08. 31				
Temperature(°C)	27 °C	Humidity (% R.H)	48 % R.H	Pressure(kPa)	100.8 kPa
Remarks	Complied				

6.10.1 Measurement procedure

The dips/interruption test is only applicable to AC mains.
The dips/interruptions were applied at zero crossing.

6.10.2 Used equipments

Equipment	Model no.	Serial no.	Makers	Next Cal. date	Used
Ultra compact simulator	UCS 500-M	V0545100858	EM TEST	2014.01.31	☐
Ultra compact simulator	UCS500M	0701-03	EM TEST	2014.06.21	☒

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6.10.3 Photographs of test setup



6.10.4 Measurement result

* 230 V, 50 Hz

Test Level (%UT)	Dip/Int. (%UT)	Duration /Period	Angle (°)	Count number	Result
0 %	100 %	0.5 Period	0 / 180	3T	A
70 %	30 %	25 Period	0	3T	A
0 %	100 %	250 Period	0	3T	C

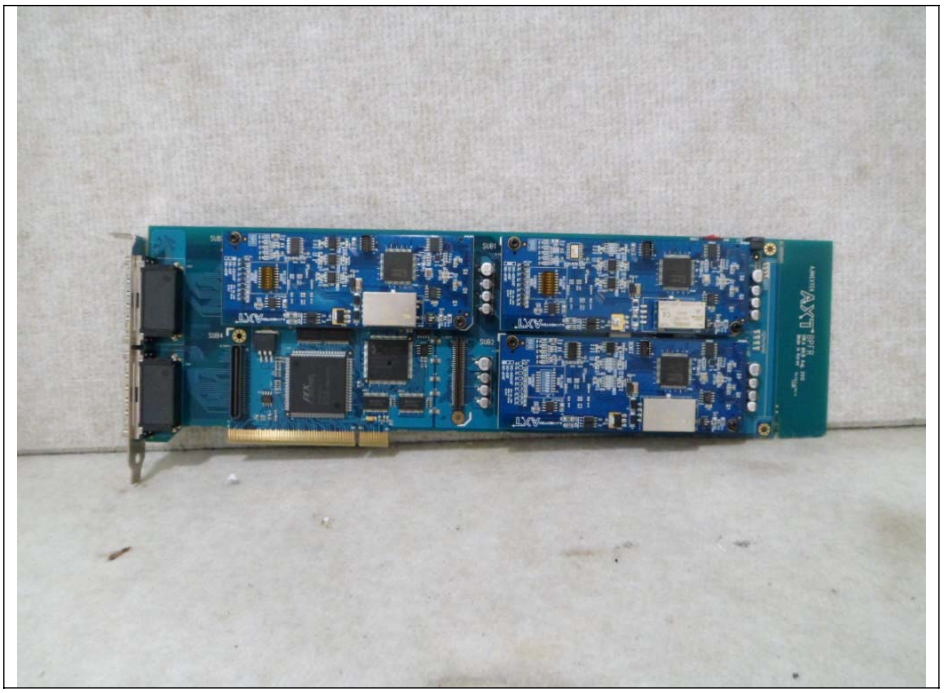
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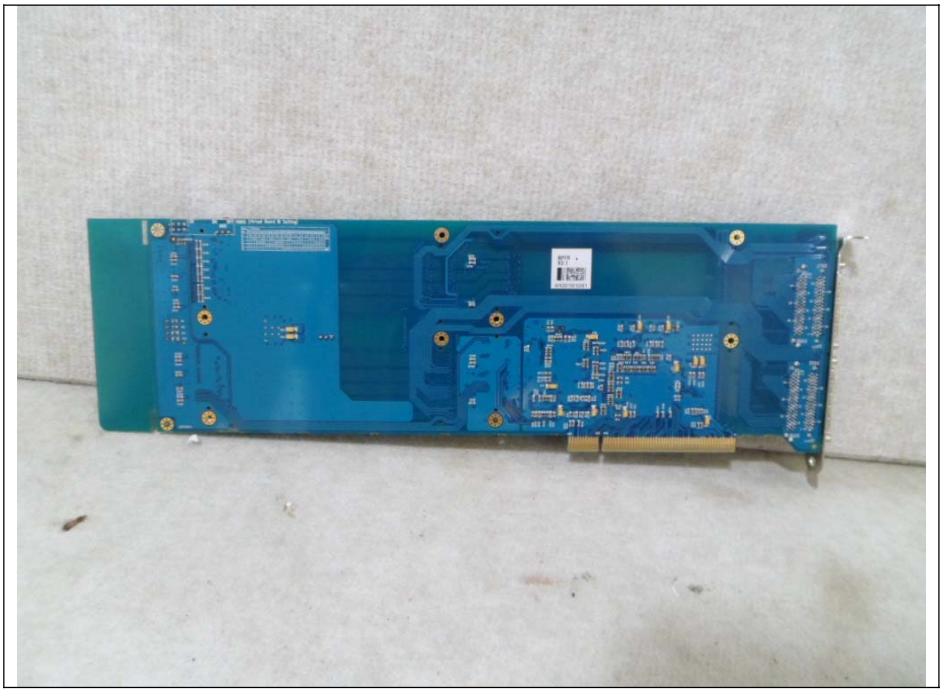
- A: There was no change of operation status during above testing. (0.5 Period, 25 Period)
- C: The power of EUT is off during the interruption test.
  After the test, EUT is getting back to normal operation. (250 Period)

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7. E.U.T. photographs

Whole

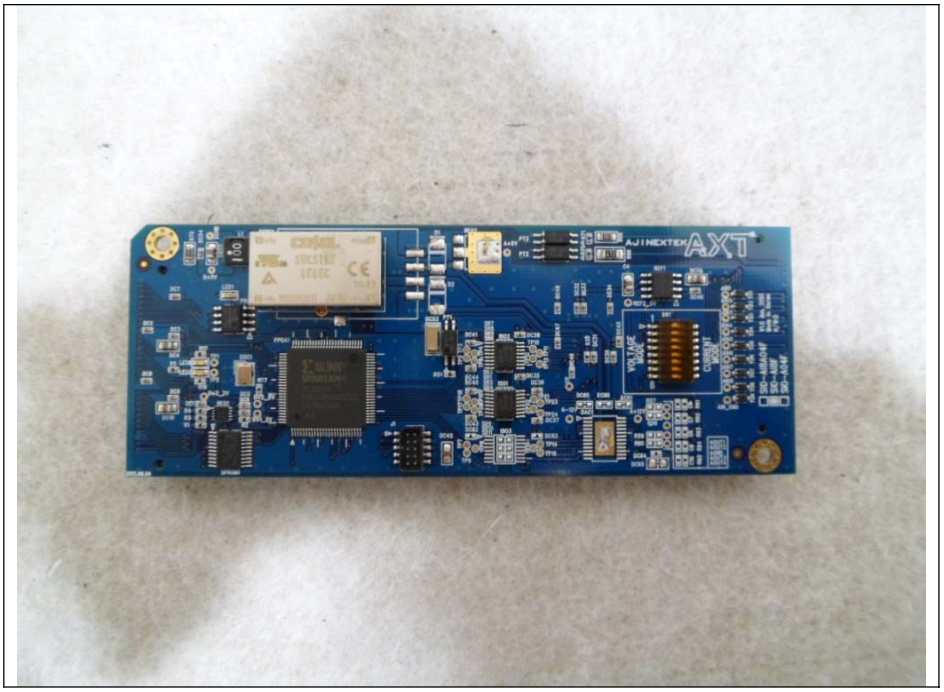




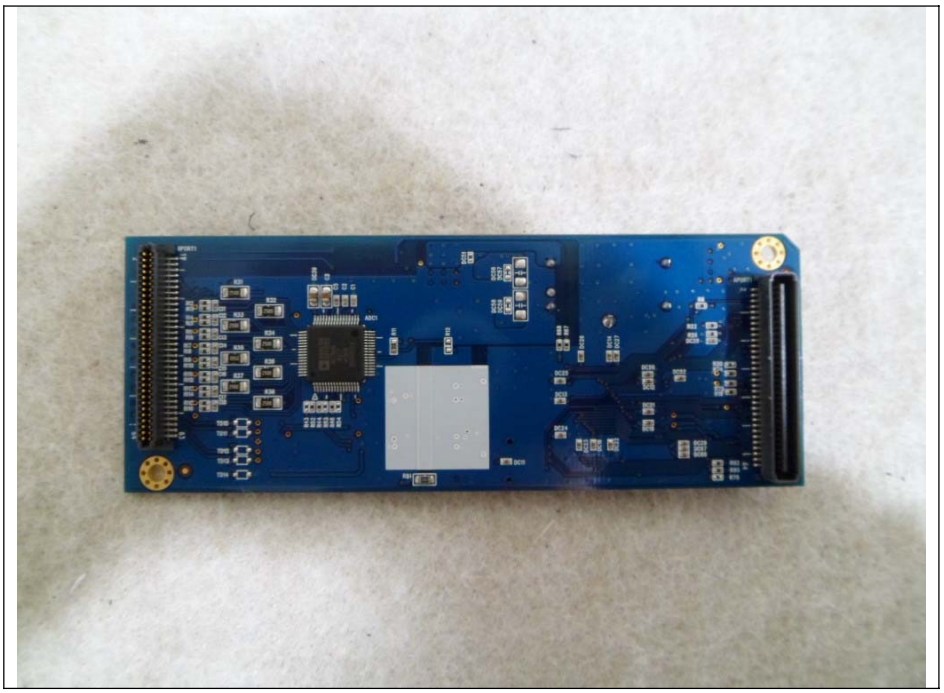
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Front View



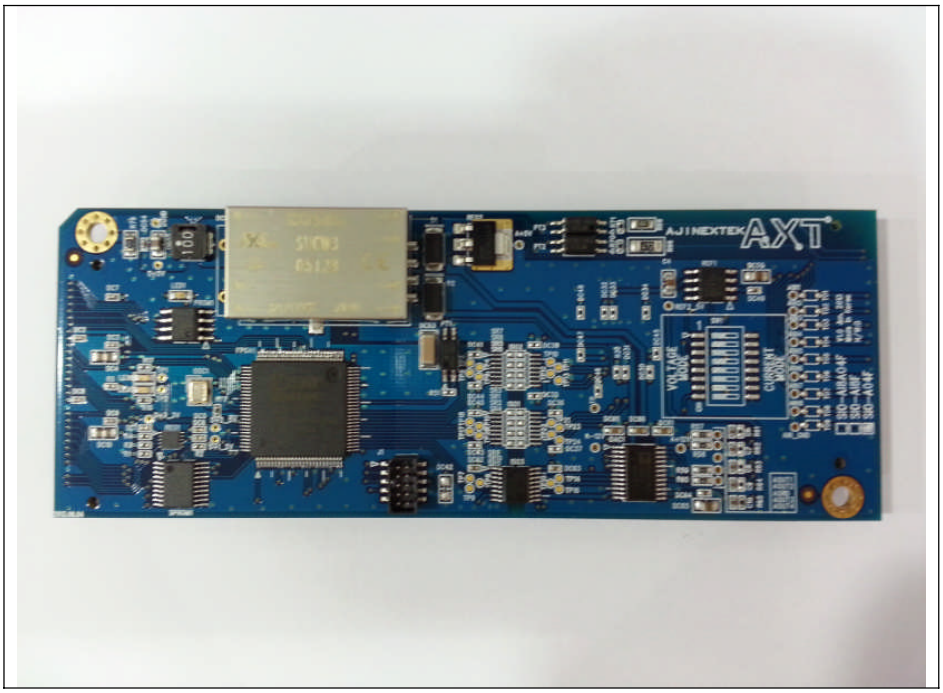
Rear View



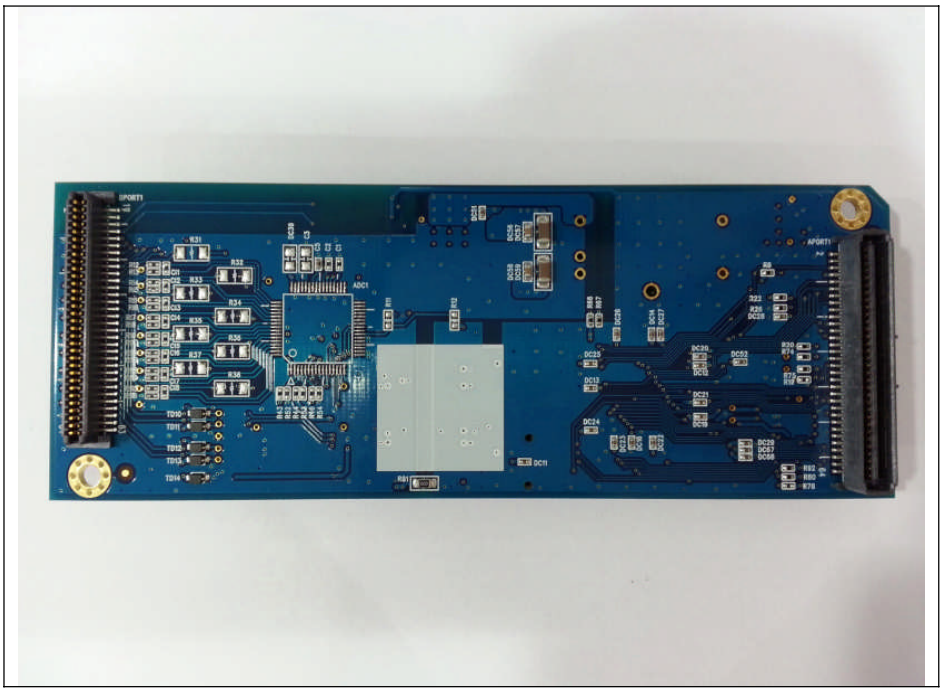
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SIO-AO4F V1.0

Front View



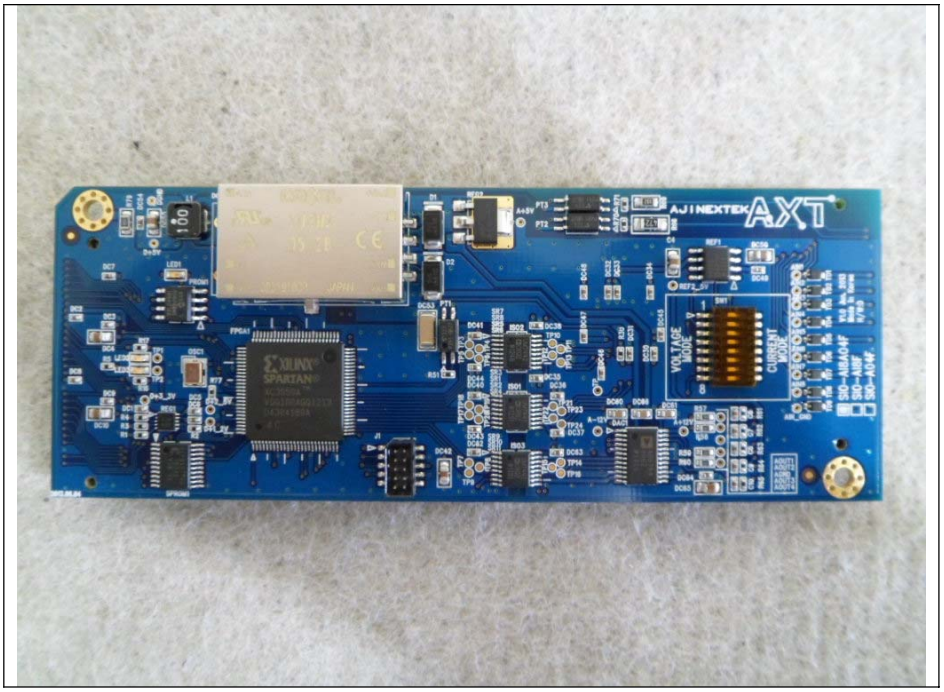
Rear View



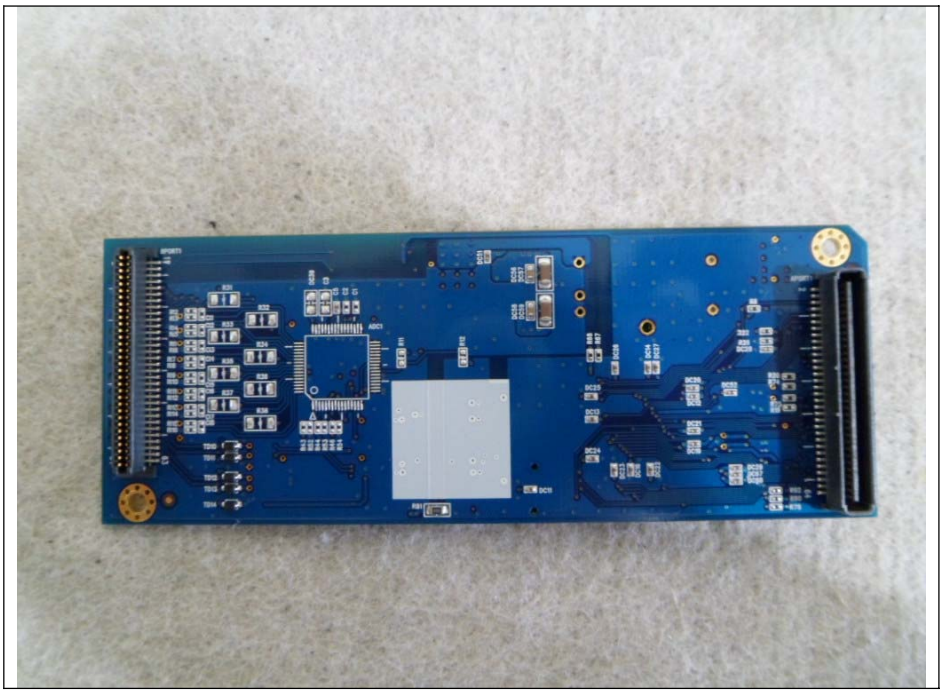
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Front View

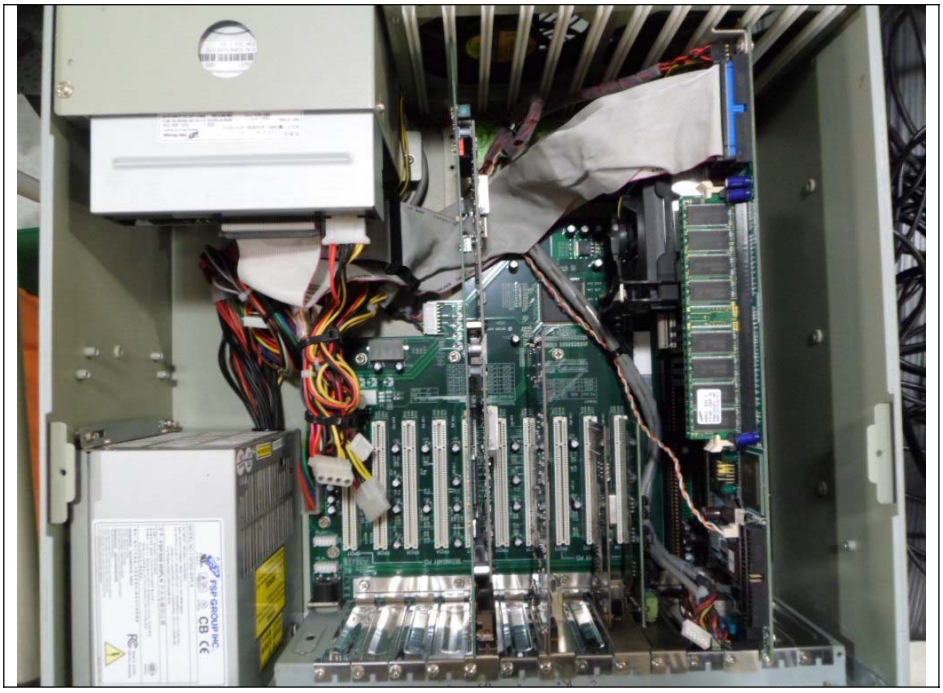


Rear View



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Inside of PC



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